

Lineage and Release Timing in Correlated Errors of Open-Weight Language Models: A Pair-Level Study

SUDHARSAN S¹, S. KANAGA SUBA RAJA^{1,2}, SHREE HARISH V^{1,3}, CHIN-SHIUH SHIEH²,
MONG-FONG HORNG², LAVANYA R¹

¹Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli 620101, India

²Research Institute of IoT Cybersecurity, Department of Electrical Engineering, National Kaohsiung University of Science and Technology, Kaohsiung 824, Taiwan, ROC

³Department of Computer Science and Engineering (Big Data Analytics), SRM Institute of Science and Technology, Tiruchirappalli 620101, India

Corresponding author: S. Kanaga Suba Raja (e-mail: kanagass@srmist.edu.in).

ABSTRACT When two open-weight language models both answer a question wrongly, how often do they choose the same wrong option, and is that more closely associated with shared ancestry or with release timing? Across 977 models from the Open LLM Leaderboard, with item-level predictions on the Massive Multitask Language Understanding (MMLU) benchmark validated item by item, two models descended from the same declared base model choose the same wrong option 13.3 percentage points more often than unrelated models of equal accuracy and release gap (95% confidence interval 10.8–15.8; mean agreement 42.3%). The association is graded, from 18.5 points for parent–child pairs to 11.5 for distant relatives, and holds across twelve pre-specified analyses, root-level inference, item-difficulty-aware null models, a chance-adjusted agreement metric and the removal of duplicate and degenerate models. Release proximity, whether of the models themselves or of their base models, shows no association of stable sign once accuracies are controlled; the primary analysis excludes same-month effects larger than 1.1 points. The analysis plan and sample rules were fixed before any pairwise outcome was computed. The study is diagnostic: it does not test ensembles, and the associations are observational in a population dominated by community fine-tunes.

INDEX TERMS Correlated errors, dyadic regression, ensemble learning, Hugging Face, large language models, model lineage, Open LLM Leaderboard

I. INTRODUCTION

WHEN several public large language models (LLMs) are asked the same hard question, they often fail together, and often in the same way. Kim *et al.* [1] document this across hundreds of open and hosted models: when two models are both wrong, they choose the same wrong option far more often than chance. For anyone who combines models—a majority vote, a routing scheme, or a review step that escalates only when models disagree—this matters directly, because agreement between members is informative only if their errors are not shared [2], [3].

Two explanations for shared errors compete. *Lineage*: a model fine-tuned from another inherits its parameters and therefore, plausibly, its blind spots. *Era*: models built in the same period draw on the same data, recipes and base models and may acquire the same mistakes without shared ancestry. The two are entangled, because a fine-tune is always released

after its base. We measure era by *release timing*: the month each model was uploaded to Hugging Face and, in a post-hoc analysis, the month its root base model was released.

This paper is a diagnostic study. It measures how shared wrong answers relate to lineage and release timing in a large population of open-weight models; it does not build or test ensembles, and whether the associations translate into better voting or routing is a separate question. Because shared errors are a property of *pairs* of models, we model agreement on wrong answers between pairs, with lineage and release-month gap as pair-level predictors, in a design whose analysis and sample rules were fixed before any outcome was computed. Decomposing per-model accuracy into family and era variance components, the obvious alternative, is too imprecise with the few families and release quarters real populations offer (Supplementary Section S1); an earlier version of this work attempted it, and its findings are withdrawn (Supple-

mentary Section S2).

A. RESEARCH QUESTIONS

- 1) **RQ1:** Among open-weight models, is sharing a lineage root associated with choosing the same wrong answer, at equal release gap and accuracy?
- 2) **RQ2:** Is release proximity associated with choosing the same wrong answer, at equal lineage status and accuracy?

B. CONTRIBUTIONS

- 1) **A pre-specified pair-level design.** Agreement on wrong answers is modelled for pairs of models, with dyadic inference for the dependence among pairs and a simulation check of whether lineage and release timing can be separated in the population studied (Section IV).
- 2) **An item-level validation protocol.** Predictions for 977 leaderboard models are checked item by item against the leaderboard's stored correctness, and two data-quality problems in the leaderboard—models listed under several names, and models that give one answer letter to almost every item—are documented and shown not to drive the results (Section III).
- 3) **Evidence on the two explanations.** Shared lineage is associated with about 13 points more agreement on wrong answers, more for closer relatives; release proximity shows no association of stable sign, bounded to about one point in the primary analysis (Section V).

Fig. 1 outlines the study.

II. RELATED WORK

Correlated errors in LLMs. Kim *et al.* [1] measure pairwise agreement on wrong answers across 349 Hugging Face models and 71 models from the Holistic Evaluation of Language Models (HELM) and regress it on pair characteristics, finding that a shared provider, a shared base architecture and similar size predict higher agreement, and that larger and more accurate models have more correlated errors. They suggest that newer models may be converging but do not test this over release time. Chen [4] shows across 67 models that the gain from routing, voting or mixture-of-agents is capped by the rate at which all models fail the same query. Kuai *et al.* [5] audit 18 models from six families for synchronized failures and reweight verifier ensembles accordingly. Goel *et al.* [6] propose chance-adjusted probabilistic agreement (CAPA), a similarity metric based on overlapping mistakes, and report that mistakes become more similar as capability increases. Kim [7] audits majority-vote gain across subsets of 30 models and finds that, once capability is controlled, shared errors are the only stable predictor of lower gain. In vision-language models, Bugaud [8] finds correlated mistakes within model families and proposes family-aware aggregation.

Null models for excess agreement. Jo, Garg and Raghavan [9] show that whether models agree “more than expected” depends on the null model of independence—accounting for item difficulty changes conclusions—and on

the population of models and items. Three estimands are in use: correlated correctness (whether two models fail the same items), conditional agreement on the wrong option (whether two models that both fail choose the same distractor [1]) and accuracy-adjusted similarity [6]. Our primary estimand is the second and our secondary the first. Our quantity of interest is a *contrast* between related and unrelated pairs rather than the level of excess agreement, so a null that shifts all pairs equally leaves it unchanged; we test this with item-difficulty-aware nulls.

Lineage and ancestry. Laufer, Oderinwale and Kleinberg [10] map the family trees of about 1.86 million Hugging Face models from declared parents and study how metadata traits are inherited. Horwitz, Shul and Hoshen [11] recover model trees from weights, and Wu *et al.* [12] infer relationships from a compact representation of model behaviour. Tamura *et al.* [13] show that declared parent-child relations improve the prediction of benchmark performance. PhyloLM [14] reconstructs phylogenies from output similarity, and Li *et al.* [15] trace the lineage of post-training datasets. None of these measures whether related models make the same errors, item by item.

Statistical tools. Crossed variance-component models estimated by restricted maximum likelihood (REML) [16]–[18] and generalizability theory [19] underlie the per-model analysis in Supplementary Section S1; Messing [20] shows that ignoring design choices in LLM pipelines understates evaluation uncertainty. Pairwise outcomes are dependent, because every model appears in many pairs; we use dyadic cluster-robust variance estimators [21]–[23] and a delete-one-root jackknife [24]. Ensemble gains depend on error diversity [25], and monoculture arguments [26], [27] treat correlated error as a risk.

Our contribution. We do not claim to be the first study of correlated errors in LLMs. Relative to Kim *et al.* [1], the closest work, the outcome is the same, but lineage is resolved per model from declared fine-tune ancestry rather than coded as shared provider or architecture, release timing is tested jointly with it at equal accuracy, and inference allows for the dependence among pairs. To our knowledge this is the first study to examine declared ancestry and release proximity jointly as correlates of agreement on wrong answers.

III. DATA AND LINEAGE

A. SOURCE AND VALIDATION

The first version of the Open LLM Leaderboard [28] evaluated thousands of open-weight models on MMLU [29] (5-shot) with the LM Evaluation Harness [30] and published per-item details, including per-option log-likelihoods. For each model we select the most complete run covering all 57 subjects and read, per item, the per-option log-likelihoods, the correct option, the leaderboard's stored correctness and, where present, an item hash. A model is accepted only if (i) all 14,042 items are present, (ii) the option with the highest log-likelihood reproduces the stored correctness on *every* item, and (iii) item order and correct options match

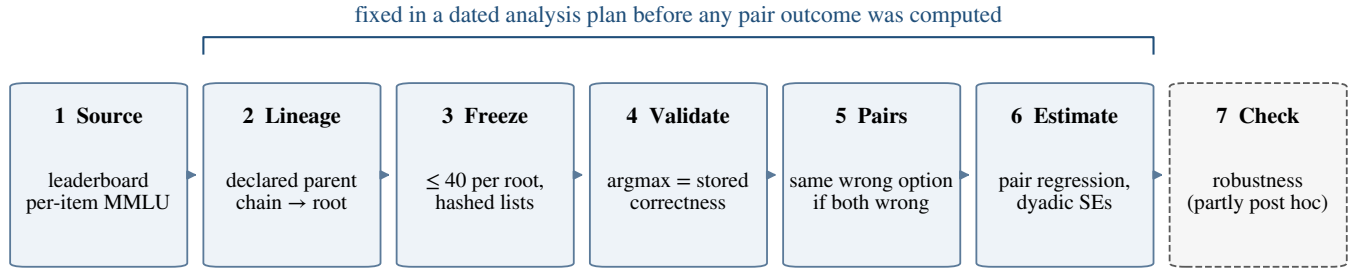


FIGURE 1: Study workflow. Steps 2–6 were fixed in a dated analysis plan before any pair outcome was computed (Section IV-D); step 7 combines pre-specified, exploratory and post-hoc checks. Counts at each step are in Fig. 2.

a reference model. Check (ii) is the decisive one: an earlier version of this work recorded the first answer option for every item, producing constant predictions that no aggregate statistic flagged (Supplementary Section S2), and that error, a misaligned item order and a misread answer key each produce thousands of disagreements. It cannot detect an error in the leaderboard’s own evaluation, such as an unsuitable prompt format for a particular model. For three spot-checked models, extracted accuracy also equals the leaderboard’s official aggregate (43.80, 70.06 and 75.56). We keep each model’s chosen option per item.

B. LINEAGE AND RELEASE TIMING

Lineage is resolved by following each model’s declared `base_model` on Hugging Face to a *root*, collapsing repository renames and a documented list of straight re-uploads and quantisations. A chain is unresolved if it passes through a merge, reaches a deleted ancestor, or ends at a fine-tune with no declared parent. Two populations are defined: *primary*, in which every link comes from a model card, and *expanded*, which also accepts a model’s own first link from the `_name_or_path` field of its configuration file when that field names an existing repository. A *strict* variant of each keeps only chains ending at a checkpoint the leaderboard lists as pretrained. *Upload month* is the Hub creation month of a model’s repository (27 months, March 2022 to July 2024); *root release month* is that of its root.

C. SAMPLES

Each population was sampled with at most 40 models per root and a fixed seed, and the samples were frozen with SHA-256 hashes before any prediction was downloaded (Fig. 2). Of the 1,018 models in the two overlapping samples, 977 passed validation; 41 were excluded and logged, and none was replaced. The analysed primary sample has 590 models from 351 roots (63 with two or more models) and 173,755 pairs; the expanded sample has 928 models from 422 roots (108) and 430,128 pairs. The largest roots are Llama-3-8B and Mistral-7B-v0.1 (39 and 37 primary models). Median accuracy is 0.415, and 247 primary models score below 0.30, close to chance for four options.

D. DATA QUALITY

Two problems in the leaderboard data affect pairwise analyses. First, some models are listed more than once: in the primary sample, 50 leaderboard entries resolve on the Hub to only 23 repositories (renamed or moved), and most such pairs give identical predictions on every item; 193 primary pairs agree on at least 99.9% of items. Second, 21 primary models choose one answer letter on at least 95% of items; all score below 0.30. Both inflate agreement for some pairs, in opposite directions for the lineage contrast. Neither was anticipated in the analysis plan; Section V-D reports the results without them.

E. HOW LINEAGE AND RELEASE TIMING OVERLAP

Fig. 3 places the primary models on a grid of root by upload month and shows what the data can identify. Most roots contribute one model (288 of 351); such models enter only different-root pairs, so the lineage contrast is identified by the 63 roots with two or more models. The fine-tunes of a root are uploaded within a few months of one another, so same-root pairs are concentrated at short gaps (711, 1,004, 259, 45 and 1 pairs at 0, 1–2, 3–5, 6–11 and 12 or more months), and gaps of a year or more are populated almost entirely by different-root pairs.

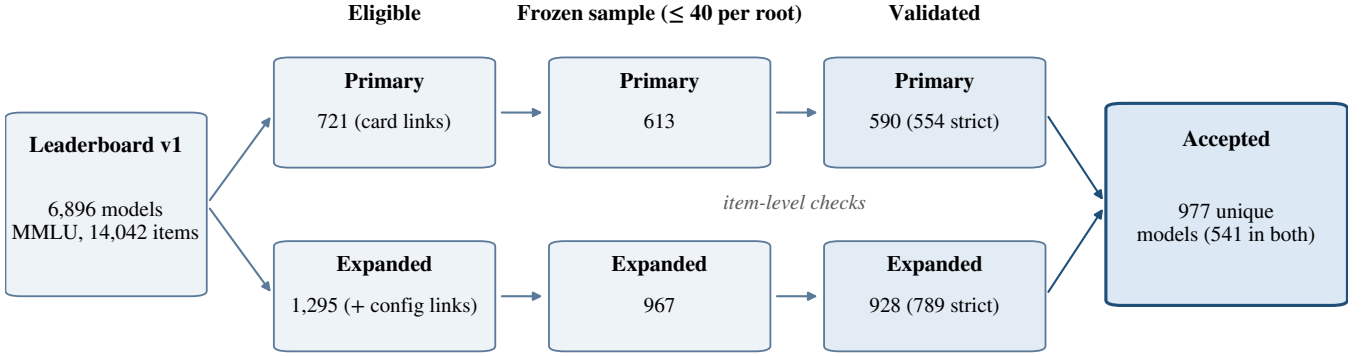
IV. STUDY DESIGN

A. OUTCOME

For models i and j , let c_{ik} be model i ’s chosen option on item k and t_k the correct option. The outcome is the conditional agreement [1]

$$a_{ij} = \frac{\sum_k \mathbf{1}[c_{ik} = c_{jk} \neq t_k]}{\sum_k \mathbf{1}[c_{ik} \neq t_k] \mathbf{1}[c_{jk} \neq t_k]}, \quad (1)$$

the probability of choosing the same wrong option given that both are wrong. Every pair contributes all items on which both models are wrong: between 1,166 and 10,820 items (median 4,109). Each item has three wrong options, so two models choosing among them independently and uniformly would give $a_{ij} = 1/3$. The outcome measures *how* two models fail when they both fail; *whether* they fail on the same items is measured by the secondary outcome, the phi coefficient of their error indicators.



Excluded and logged, none replaced: 33 no complete run, 7 repository missing, 1 unreadable (of 1,018 frozen)

FIGURE 2: Populations, samples and validation. The primary population accepts only lineage links declared on model cards; the expanded population also accepts a model's first link from its configuration file. The two frozen samples overlap, so the 977 accepted models are counted once. Strict: root verified as a pretrained checkpoint. Item-level checks: all 14,042 items present; the highest-scoring option reproduces the stored correctness on every item; item order and correct options match a reference model.

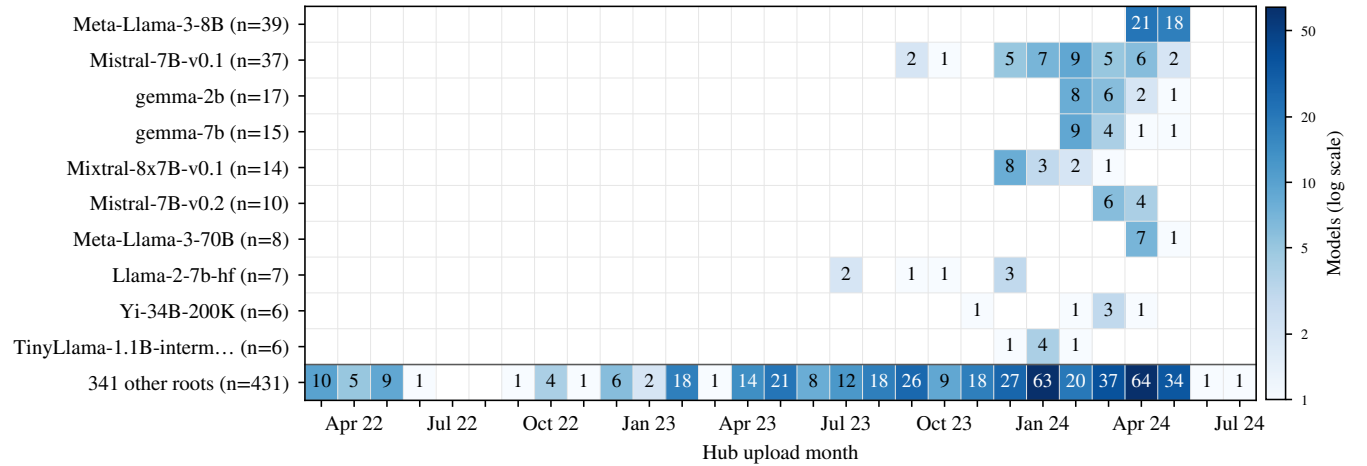


FIGURE 3: Lineage roots by upload month (primary sample, 590 models). Each cell counts the models of one root uploaded in one month (log color scale). The 10 largest roots are shown individually; the bottom row pools the other 341 roots, most of which contribute one model. Fine-tunes of a root cluster within a few months, so same-root pairs occur mainly at short release gaps.

B. PAIR MODEL

The pre-specified model is

$$a_{ij} = b_0 + b_L s_{ij} + \sum_h b_{E,h} \mathbf{1}[|m_i - m_j| \in h] + \gamma_1(\text{acc}_i + \text{acc}_j) + \gamma_2|\text{acc}_i - \text{acc}_j| + \varepsilon_{ij}, \quad (2)$$

estimated by ordinary least squares (OLS), where s_{ij} indicates a shared lineage root, m_i is the upload month, and h ranges over gap bins of 0, 1–2, 3–5 and 6–11 months, with 12 or more months as reference. We call b_L the *shared-root coefficient* and $b_{E,0}$ the *same-month coefficient*: the difference in agreement between pairs uploaded in the same month and pairs uploaded 12 or more months apart, at equal lineage status and accuracies. Coefficients are differences in probability; in the text we report them in percentage points. Both are conditional associations, not causal effects. Fig. 4 illustrates the two predictors: every pair carries one value of each, so

same-root pairs occur at several gaps and same-month pairs occur with and without a shared root.

The accuracy terms are pre-specified because two models of similar accuracy tend to fail on similar items, and the items strong models fail are the hardest, which may have more attractive distractors. They are not neutral: if lineage or release timing affects agreement partly through accuracy, adjustment removes that part, so we also report the model without them.

C. INFERENCE

Every model enters many pairs, so pair residuals are dependent. The pre-specified standard errors (SEs) are dyadic cluster-robust [21]–[23]: with x_p the covariates and $\hat{\varepsilon}_p$ the

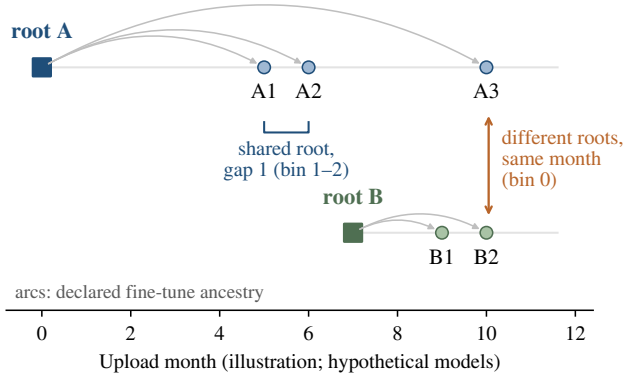


FIGURE 4: The two pair-level predictors (hypothetical models). Squares are lineage roots, circles fine-tunes at their upload month; arcs show declared ancestry. A1 and A2 share a root and were uploaded one month apart (gap bin 1–2); A3 and B2 have different roots but the same upload month (gap bin 0).

residual of pair p ,

$$\hat{V}(\hat{\mathbf{b}}) = (X^\top X)^{-1} \left(\sum_{p,q: p \cap q \neq \emptyset} \hat{e}_p \hat{e}_q \mathbf{x}_p \mathbf{x}_q^\top \right) (X^\top X)^{-1}, \quad (3)$$

where $p \cap q \neq \emptyset$ means that pairs p and q share a model (including $p = q$). Because fine-tunes of one root may err alike beyond what sharing a model captures, we also use two procedures that treat the root as the unit of dependence: root-level dyadic clustering (Eq. (3) with “share a root”) and a delete-one-root jackknife, which refits the model G times, omitting every pair that involves one of the G roots, and estimates the variance as $\frac{G-1}{G} \sum_r (\hat{\mathbf{b}}_{(r)} - \hat{\mathbf{b}})^2$. The jackknife gives the widest intervals for both coefficients in every sample, so we report it as the main interval.

D. ANALYSIS PLAN AND DEVIATIONS

The plan, written before any pair outcome was computed, fixed the outcome, Eq. (2), the model-level SEs, the phi outcome as secondary, a tree-distance lineage predictor as secondary, the sample rules and a simulation check (Section IV-E). Three dated amendments followed, each before the quantities it concerns were computed: (1) the source moved from the second to the first version of the leaderboard, whose item-level files are not access-gated; (2) the populations were defined as above and the cap was set to 40 models per root; (3) after the accuracies of the first 443 validated models showed many near-chance models, two sensitivity analyses were added: *S1* adds the total-variation distance $\frac{1}{2} \sum_o |\pi_i(o) - \pi_j(o)|$ between the two models’ distributions π over chosen answer options as a covariate, because two models favouring the same letters agree more for that reason alone, and *S2* keeps models with accuracy at least 0.30. Amendment 3 is the one decision informed by data, though by per-model accuracies rather than pair outcomes. The plan was committed to the project repository but not deposited with an

TABLE I: Status of each analysis. Pre-spec.: in the analysis plan. Amend.: dated amendment before any pair outcome. Explor.: not in the plan, run with or after the pre-specified analyses. Post hoc: added in response to review. WLS: weighted least squares; CAPA: chance-adjusted agreement.

Analysis	Status
Pair model, Eq. (2); model-level dyadic SEs; phi outcome	Pre-spec.
Simulation check and its criteria	Pre-spec.
Leaderboard version 1; populations; cap of 40	Amend. 1–2
S1 (option distribution), S2 (accuracy ≥ 0.30)	Amend. 3
Root-level SEs; WLS; option-chance outcome; size controls	Explor.
Without accuracy terms; item-difficulty nulls; CAPA	Post hoc
Root release month; duplicate, degenerate and config checks	Post hoc
500-replicate simulation audit; cap-15 subset	Post hoc
Tree-distance predictor (in the plan, run after review)	Post hoc
Replication on leaderboard version 2 (access-gated)	Not done

independent registry, so we call it pre-specified rather than pre-registered. TABLE I gives the status of every analysis.

E. SIMULATION CHECK

Because same-root pairs are concentrated at short gaps (Fig. 3), a release-time effect could leak into the lineage coefficient or vice versa. Before any pair outcome was computed, we simulated item-level choices on the actual roster. Each model answers correctly with probability $\text{logit}^{-1}(\theta_i - d_k)$, with $\theta_i \sim \mathcal{N}(0.3, 1)$ and $d_k \sim \mathcal{N}(0, 1)$; when wrong, it picks a root-specific preferred option with probability λ_L , a month-specific preferred option (persisting from month to month with probability 0.8) with probability λ_E , an option preferred by all models with probability $\lambda_A = 0.15$, and otherwise a uniform wrong option. Eq. (2) with dyadic SEs was refitted to each simulated data set. The criteria were: false rejection under no effect at most 10%; *leakage*, the rejection rate of the wrong coefficient under a single true effect, at most 10%; and power at least 80% at $\lambda = 0.05$. With 20 replicates per scenario, the cap of 40 met all three. A post-hoc audit with 500 replicates (Supplementary Section S3) confirms the null and power criteria but finds leakage of 10.8% (95% CI 8.4–13.8) into the lineage coefficient at the strongest release-time effect, $\lambda_E = 0.10$, statistically indistinguishable from the 10% criterion. The implied bias on the observed lineage estimate is about 3×10^{-4} .

V. RESULTS

A. THE RAW PATTERN

Fig. 5 shows the finding before any modelling. At every release gap at which both kinds of pair occur, same-root pairs choose the same wrong option on 71–75% of jointly wrong items and different-root pairs on 40–45% (overall 73.1% and 42.0%, mean 42.3%). Among different-root pairs, agreement declines slowly with the gap, from 0.45 in the same month to 0.37 at a year or more. Kim *et al.* report the same mean for

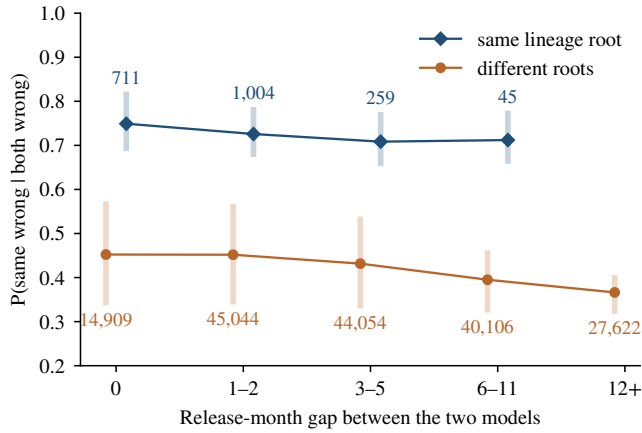


FIGURE 5: Agreement on wrong answers by release-month gap (primary sample, no adjustment). Points: mean of a_{ij} ; shaded bars: interquartile range across pairs; labels: number of pairs. Bins with fewer than 20 pairs are omitted.

a later leaderboard snapshot [1], but on a different question set with mostly ten answer options, so the coincidence is not evidence of validity.

B. PRIMARY RESULT AND PRE-SPECIFIED ANALYSES

In the primary analysis, sharing a lineage root is associated with a 13.3-point higher probability of choosing the same wrong option (jackknife 95% CI 10.8–15.8; model-level CI 11.8–14.8). Two models uploaded in the same month agree 0.4 points more than two uploaded a year or more apart (jackknife CI -0.5 to $+1.3$). Across the twelve pre-specified analyses (TABLE II, Fig. 6) the shared-root coefficient ranges from 11.7 to 13.4 points. The same-month coefficient is positive when the answer-option distribution is controlled (S1: $+1.0$, CI $+0.05$ to $+1.95$) and negative among models above 0.30 accuracy (S2: -1.5 , model-level CI -2.7 to -0.2 , jackknife CI -3.0 to $+0.1$): its sign is not stable. The other gap bins show the same pattern. In plain terms, when two fine-tunes of one base model both fail an item, they pick the same distractor about three times in four; two unrelated models do so about two times in five.

C. ADJUSTING FOR ACCURACY

Accuracy can mediate both associations: fine-tunes of one root tend to have similar accuracies, and so do models of one period. Without the accuracy terms (Fig. 7), the shared-root coefficient is 28.3 points and the release-gap coefficients are graded, from 8.7 points in the same month to 2.9 at 6–11 months. With them, the shared-root coefficient is 13.3 points and every release-gap coefficient is within 0.6 points of zero in both populations. If accuracy is a mediator, the unadjusted coefficients are total associations and the adjusted ones the part not running through accuracy; if accuracy and the predictor share a cause, adjustment removes a confounded component. The data cannot distinguish these readings. Under either, the predictors differ in the same way: once both accuracies are

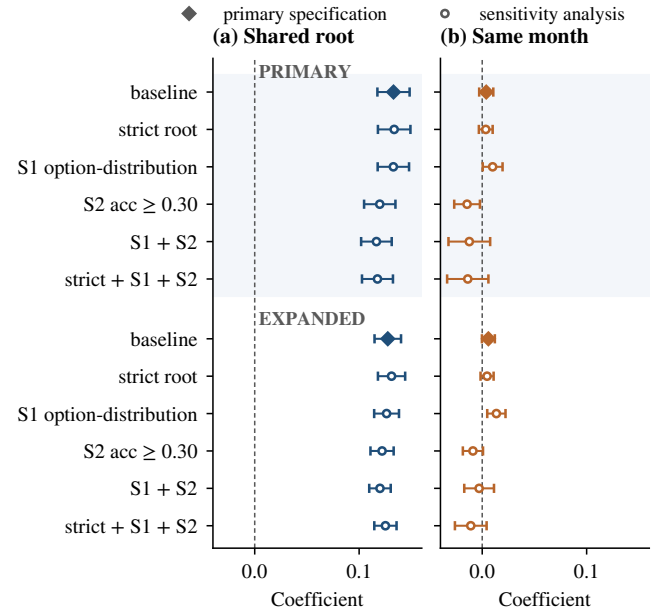


FIGURE 6: Shared-root and same-month coefficients in the twelve pre-specified analyses (95% CIs, model-level dyadic SEs), on one horizontal scale. Diamonds: the primary specification of each population.

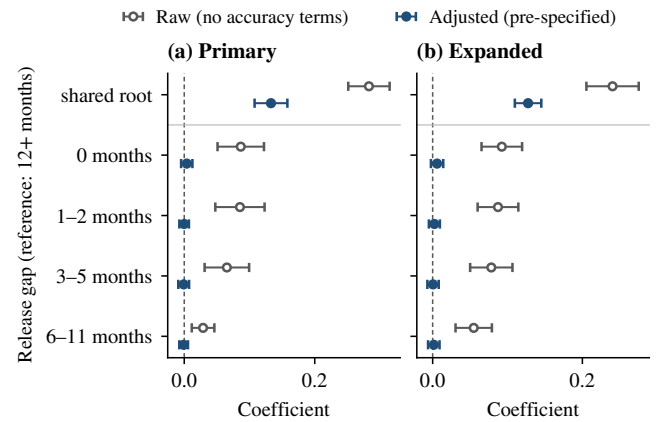


FIGURE 7: Coefficients with and without the accuracy terms (delete-one-root jackknife 95% CIs). Release-gap bins are relative to 12 or more months. Adding the accuracy terms halves the shared-root coefficient and removes the graded release-gap association in both populations.

known, release proximity adds nothing of stable sign, while a shared root still adds about 13 points.

D. ROBUSTNESS

TABLE III collects every robustness check for the primary sample; detailed tables for all samples are in Supplementary Section S4.

Inference. Root-level procedures widen the shared-root interval by 10–26% (root-level dyadic) and 39–63% (jackknife) across the four samples, and every interval stays above 0.09. Only the S2 same-month coefficient changes conclusion: its

TABLE II: Pre-specified analyses. Coefficients of Eq. (2) with 95% CIs from model-level dyadic SEs; same month is the 0-month gap bin relative to 12 or more months, and p refers to it. Strict: root verified as pretrained; S1: answer-option-distribution covariate; S2: models with accuracy ≥ 0.30 .

Analysis	Models	Pairs	Shared root	Same month	p (same month)
Primary	590	173,755	+0.133 [+0.118, +0.148]	+0.0039 [−0.0029, +0.0106]	0.263
Primary, strict	554	153,181	+0.134 [+0.118, +0.149]	+0.0034 [−0.0032, +0.0100]	0.313
Primary + S1 (option distribution)	590	173,755	+0.133 [+0.118, +0.148]	+0.0100 [+0.0005, +0.0195]	0.039
Primary, S2 (acc ≥ 0.30)	343	58,653	+0.120 [+0.105, +0.135]	−0.0145 [−0.0268, −0.0022]	0.021
Primary + S1, S2	343	58,653	+0.117 [+0.102, +0.131]	−0.0124 [−0.0323, +0.0076]	0.225
Primary, strict + S1, S2	319	50,721	+0.118 [+0.103, +0.132]	−0.0138 [−0.0336, +0.0059]	0.170
Expanded	928	430,128	+0.127 [+0.115, +0.140]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded, strict	789	310,866	+0.131 [+0.118, +0.144]	+0.0047 [−0.0016, +0.0109]	0.145
Expanded + S1 (option distribution)	928	430,128	+0.126 [+0.114, +0.138]	+0.0136 [+0.0048, +0.0223]	0.002
Expanded, S2 (acc ≥ 0.30)	589	173,166	+0.122 [+0.111, +0.133]	−0.0089 [−0.0185, +0.0008]	0.071
Expanded + S1, S2	589	173,166	+0.120 [+0.110, +0.130]	−0.0030 [−0.0172, +0.0113]	0.683
Expanded, strict + S1, S2	481	115,440	+0.125 [+0.115, +0.136]	−0.0110 [−0.0261, +0.0042]	0.157

jackknife interval includes zero.

Outcome definition. Weighting pairs by their number of jointly wrong items (weighted least squares, WLS) gives 14.2 points for lineage. Subtracting the agreement expected from each model’s distribution over answer options gives 12.5 points. Two outcomes that measure *which items* fail together—the phi coefficient and a CAPA-style agreement computed from chosen options—show strong lineage associations and a small positive same-month association (+1.8 and +1.3 points); for CAPA and for Rasch co-failure (below) it disappears among models above 0.30 accuracy. Release proximity may thus relate weakly to which items fail, but not to which wrong option is chosen.

Item-difficulty-aware nulls. Null N1 sets the expected agreement on item k to $q_k = \sum_o p_k(o)^2$, where $p_k(o)$ is the share of wrong models choosing wrong option o (N1a: all models; N1b: roots weighted equally); null N2 fits a Rasch model and measures co-failure beyond its prediction. Item-level distractor popularity accounts for almost all of the *average* agreement, which falls from 0.423 to 0.002 (N1a) and 0.019 (N1b), consistent with [9]; the lineage *contrast* is unchanged at 13.3 points.

Controls, timing measure and data quality. Adding model size and a same-architecture indicator leaves 12.6 points; lineage and architecture are nested here, so this partitions one association. Measuring release timing by the roots’ release months leaves the lineage coefficient at 13.1 points and gives a same-month coefficient of −0.1 points (CI −1.2 to +1.0); across samples it is never significantly positive. Removing the degenerate models and collapsing identical-prediction clusters gives 12.8 points; keeping only models whose configuration file agrees with their declared root gives 13.5; the cap-15 subset of the sample gives 14.6.

Equivalence bounds. The 90% jackknife interval for the pre-specified same-month coefficient excludes associations larger than 1.1 points in the primary sample and 1.3 in the expanded sample (2.0–2.8 in the S2 samples), against 13.3 for lineage. These bounds were not specified in advance.

Independent check of lineage. For 191 primary models, the configuration file names another Hub repository; follow-

ing that link reaches the declared root in 142 cases (74%), another root in 48 and none in 1. The disagreements mix declaration errors, such as continued-pretraining models declared as their own roots, with configuration files copied from other models.

E. LINEAGE DOSE-RESPONSE

If shared errors are inherited, closer relatives should agree more. Replacing the shared-root indicator by tree distance gives a monotone gradient (TABLE III): 18.5 points for parent–child pairs (177 pairs), 13.6 at distance two (699, mostly siblings) and 11.5 for more distant relatives (1,112). Closer relatives are also uploaded closer together and often by the same developer, so the gradient is consistent with inheritance but does not isolate it.

VI. DISCUSSION

What the results say. In this population, which base model a fine-tune descends from is strongly associated with which wrong answers it gives: related pairs agree on wrong answers about 13 points more often, a 31% increase over the mean, and more the closer the relation. When a model was released adds nothing of stable sign once accuracies are known. This is consistent with Kim *et al.*’s finding that shared provider and architecture predict agreement [1], and refines it: with lineage taken from declared ancestry and timing measured directly, the association runs through lineage.

What they do not say. The association is observational. Fine-tunes of one base share its parameters, its pretraining data and often similar post-training data; the design does not separate these routes or show that fine-tuning causes inherited errors. Nor does it show that release timing is irrelevant: the pretraining era of a base model is shared by its fine-tunes and therefore nested in lineage here, contemporaneous models agree more partly because they are similarly accurate, and timing may relate weakly to which items fail together.

Why lineage and not release timing. A fine-tune starts from its base model’s weights and is typically trained on far less data than its base; much of the base model’s knowledge, and its gaps, plausibly survive. Models uploaded in the same month share only what was common at the time—popular

TABLE III: Robustness checks (primary sample; 95% CIs from the SE type shown). Same month: 0-month gap bin relative to 12 or more months. ^aDifferent outcome scale (correlation, chance-adjusted agreement or co-failure per item). ^bTree distance replaces the shared-root indicator: the number of declared links between the two models through their nearest common ancestor.

Check	Status	SE	Shared root	Same month
Pre-specified model	Pre-spec.	model	+0.133 [+0.118, +0.148]	+0.0039 [−0.0029, +0.0106]
root-level dyadic SEs	Explor.	root	+0.133 [+0.114, +0.152]	+0.0039 [−0.0041, +0.0118]
delete-one-root jackknife	Explor.	jackknife	+0.133 [+0.108, +0.158]	+0.0039 [−0.0048, +0.0125]
Without accuracy terms	Post hoc	jackknife	+0.283 [+0.251, +0.315]	+0.0866 [+0.0510, +0.1222]
WLS (weight: jointly wrong items)	Explor.	model	+0.142 [+0.125, +0.159]	+0.0117 [+0.0016, +0.0217]
Minus option-distribution chance	Explor.	model	+0.125 [+0.111, +0.140]	−0.0018 [−0.0046, +0.0011]
Phi of error indicators ^a	Pre-spec.	model	+0.170 [+0.149, +0.191]	+0.0184 [+0.0097, +0.0272]
CAPA, chance-adjusted agreement ^a	Post hoc	jackknife	+0.194 [+0.153, +0.234]	+0.0126 [+0.0020, +0.0232]
N1a: minus item distractor null	Post hoc	jackknife	+0.133 [+0.107, +0.159]	+0.0040 [−0.0047, +0.0127]
N1b: as N1a, roots weighted equally	Post hoc	jackknife	+0.133 [+0.108, +0.159]	+0.0038 [−0.0049, +0.0124]
N2: co-failure over Rasch null ^a	Post hoc	jackknife	+0.040 [+0.032, +0.048]	+0.0036 [+0.0012, +0.0060]
+ size and architecture controls	Explor.	model	+0.126 [+0.110, +0.142]	−0.0071 [−0.0027, +0.0169]
Root release month for timing	Post hoc	jackknife	+0.131 [+0.108, +0.154]	+0.0007 [−0.0115, +0.0101]
Without duplicate and degenerate models	Post hoc	jackknife	+0.128 [+0.106, +0.150]	−0.0024 [−0.0081, +0.0034]
Config-consistent lineage only	Post hoc	jackknife	+0.135 [+0.108, +0.161]	+0.0002 [−0.0069, +0.0072]
Cap-15 subset of the sample	Post hoc	jackknife	+0.146 [+0.118, +0.174]	+0.0048 [−0.0045, +0.0141]
Lineage tree distance 1	Post hoc ^b	jackknife	+0.185 [+0.157, +0.213]	—
Lineage tree distance 2	Post hoc ^b	jackknife	+0.136 [+0.117, +0.155]	—
Lineage tree distance 3+	Post hoc ^b	jackknife	+0.115 [+0.103, +0.128]	—

base models, datasets and recipes—and in this population that commonality is already captured by lineage and accuracy. These are hypotheses consistent with the results, not mechanisms the design tests.

Implications. The results suggest, as a hypothesis for practitioners, that drawing ensemble members from different base models may reduce shared wrong answers more than drawing them from one base at different times. We did not test this. The effect on ensemble accuracy depends on how members are combined [4], and capability-controlled audits find that shared errors predict majority-vote gain only modestly [7].

VII. LIMITATIONS

- 1) **Population and design.** The associations are observational, in a self-selected population dominated by community fine-tunes of a few dozen bases from 2022 to 2024. They may not hold for frontier or closed models, or for separately pretrained models released at the same time.
- 2) **Lineage measurement.** Declared parents can be wrong or missing. Random misclassification would bias the lineage contrast towards zero; misclassification related to how models err could bias it either way. Excluding undeclared models shifts the population towards documented fine-tunes. Estimates agree across the primary, expanded and strict populations (12.7–13.4 points) and in config-consistent models, but lineage was not verified from weights.
- 3) **Release-timing measurement.** Upload and root release months are proxies for when training data were assembled, and for same-root pairs the pretraining era is shared by construction, so pretraining era cannot be separated from lineage.
- 4) **Benchmark and pipeline.** One benchmark with four options is used, and all predictions come from the

leaderboard’s evaluation pipeline. Model-specific configuration errors cannot be ruled out; 339 of 977 models score below 0.30, which the S1 and S2 analyses address only in part. Contamination could make agreement partly reflect shared memorization.

- 5) **Outcome scope.** The outcome counts only items both models get wrong; it is not a measure of ensemble performance, and its level depends on the null model and population [9].
- 6) **Identification.** Same-root pairs are almost all within a year of each other, so the data cannot show whether the lineage association persists between fine-tunes released far apart. The contrast comes from 63 roots, two of which contribute 39 and 37 models.
- 7) **Design check and plan.** The simulation check narrowly missed its leakage criterion at the strongest simulated release-time effect, and model-level SEs were 4–6% too small in simulation; we therefore report root-level intervals. The analysis plan was not registered externally, and several analyses are post hoc (TABLE I).

VIII. CONCLUSION

Among open-weight models on the first Open LLM Leaderboard, two models that share a declared lineage root choose the same wrong MMLU option, when both are wrong, about 13 percentage points more often than unrelated models of equal accuracy and release gap, and closer relatives agree more. The estimate is stable across pre-specified analyses, root-level inference, alternative outcomes, item-difficulty-aware nulls and cleaned samples. Net of accuracy, release proximity—of the models or of their base models—shows no association of stable sign with the choice of wrong option, and in the primary analysis any such association is smaller than about one point. The next step is to test whether lineage diversity improves ensemble, voting or review performance

on other benchmarks, evaluation pipelines and independently pretrained models.

DATA AVAILABILITY

Code, frozen model lists with SHA-256 hashes, Hugging Face metadata caches, validation reports and all analysis outputs are available at <https://github.com/Sudharsanselv-araj/Identifiable-Variance-Decomposition-of-Correlated-Errors-in-Foundation-Models>; the README lists the commands that regenerate every reported number, table and figure from these outputs. The leaderboard's per-item details and metadata declare no licence, so the derived item-level answer files and three copied metadata fields are not redistributed; released scripts regenerate them from the public source and verify them against the recorded hashes. The analysis plan and its amendments are in the repository. Supplementary material accompanies this article.

ACKNOWLEDGMENT

This work was supported in part by the Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli, India, and in part by the Research Institute of IoT Cybersecurity, National Kaohsiung University of Science and Technology, Kaohsiung, Taiwan.

Use of AI-generated content. The authors used Claude (Anthropic; model Claude Opus 5.5, accessed through Claude Code) [31] in October 2026, with substantial use. It wrote and debugged the analysis, simulation, data-extraction and figure/table-generation code, and drafted the text of every section of this revised manuscript (Sections I–VIII) and of the supplementary material. The authors directed the study, approved each analysis decision and analysis-plan amendment, reviewed and edited the text, and take full responsibility for the content; every reported number is produced by the released code from the committed outputs.

REFERENCES

- [1] E. Kim, A. Garg, K. Peng, and N. Garg, "Correlated errors in large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 267, pp. 30038–30066, 2025.
- [2] T. G. Dietterich, "Ensemble methods in machine learning," in *Proc. Int. Workshop Multiple Classifier Syst.*, LNCS 1857, 2000, pp. 1–15.
- [3] L. I. Kuncheva and C. J. Whitaker, "Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy," *Mach. Learn.*, vol. 51, no. 2, pp. 181–207, 2003.
- [4] J. L. Chen, "When does combining language models help? A co-failure ceiling on routing, voting, and mixture-of-agents across 67 frontier models," *arXiv preprint arXiv:2606.27288*, 2026.
- [5] C. Kuai et al., "A statistical framework for auditing behavioral dependence and induced bias in LLM judges," *arXiv preprint arXiv:2604.07650v2*, 2026.
- [6] S. Goel, J. Struber, I. A. Auzina, K. K. Chandra, P. Kumaraguru, D. Kiela, A. Prabhu, M. Bethge, and J. Geiping, "Great models think alike and this undermines AI oversight," *arXiv preprint arXiv:2502.04313*, 2025.
- [7] D. Kim, "Are diversity metrics measuring diversity? A capability-controlled audit of majority-vote gain in LLM ensembles," *arXiv preprint arXiv:2607.20768*, 2026.
- [8] Z. Bugaud, "Hidden clones: Exposing and fixing family bias in vision-language model ensembles," *arXiv preprint arXiv:2603.17111*, 2026.
- [9] N. Jo, N. Garg, and M. Raghavan, "The subjectivity of monoculture," *arXiv preprint arXiv:2602.24086*, 2026.
- [10] B. Laufer, H. Oderinwale, and J. Kleinberg, "Anatomy of a machine learning ecosystem: 2 million models on Hugging Face," *arXiv preprint arXiv:2508.06811*, 2025.
- [11] E. Horowitz, A. Shul, and Y. Hoshen, "Unsupervised model tree heritage recovery," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [12] Z. Wu, H. Zhao, Z. Wang, J. Guo, Q. Wang, and B. He, "LLM DNA: Tracing model evolution via functional representations," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2026.
- [13] T. Tamura, T. Yano, M. Enomoto, and M. Oyamada, "Can a crow hatch a falcon? Lineage matters in predicting large language model performance," *arXiv preprint arXiv:2504.19811*, 2025.
- [14] N. Yax, P.-Y. Oudeyer, and S. Palminteri, "PhyloLM: Inferring the phylogeny of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [15] Y. Li et al., "Tracing the roots: A multi-agent framework for uncovering data lineage in post-training LLMs," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguist. (Volume 1: Long Papers)*, 2026, pp. 9606–9625.
- [16] H. D. Patterson and R. Thompson, "Recovery of inter-block information when block sizes are unequal," *Biometrika*, vol. 58, no. 3, pp. 545–554, 1971.
- [17] D. A. Harville, "Maximum likelihood approaches to variance component estimation and to related problems," *J. Amer. Statist. Assoc.*, vol. 72, no. 358, pp. 320–338, 1977.
- [18] S. R. Searle, G. Casella, and C. E. McCulloch, *Variance Components*. New York, NY, USA: Wiley, 1992.
- [19] R. L. Brennan, *Generalizability Theory*. New York, NY, USA: Springer, 2001.
- [20] S. Messing, "Hidden measurement error in LLM pipelines distorts annotation, evaluation, and benchmarking," *arXiv preprint arXiv:2604.11581*, 2026.
- [21] M. Fafchamps and F. Gubert, "The formation of risk sharing networks," *J. Develop. Econ.*, vol. 83, no. 2, pp. 326–350, 2007.
- [22] A. C. Cameron, J. B. Gelbach, and D. L. Miller, "Robust inference with multiway clustering," *J. Bus. Econ. Statist.*, vol. 29, no. 2, pp. 238–249, 2011.
- [23] P. M. Aronow, C. Samii, and V. A. Assenova, "Cluster-robust variance estimation for dyadic data," *Political Anal.*, vol. 23, no. 4, pp. 564–577, 2015.
- [24] B. Efron, *The Jackknife, the Bootstrap and Other Resampling Plans*. Philadelphia, PA, USA: SIAM, 1982.
- [25] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [26] J. Kleinberg and M. Raghavan, "Algorithmic monoculture and social welfare," *Proc. Nat. Acad. Sci. USA*, vol. 118, no. 22, 2021.
- [27] R. Bommasani, K. A. Creel, A. Kumar, D. Jurafsky, and P. Liang, "Picking on the same person: Does algorithmic monoculture lead to outcome homogenization?" in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022.
- [28] E. Beeching et al., "Open LLM Leaderboard (2023–2024)," Hugging Face, 2023. [Online]. Available: https://huggingface.co/spaces/open-llm-leaderboard-old/open_llm_leaderboard
- [29] D. Hendrycks et al., "Measuring massive multitask language understanding," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.
- [30] L. Gao et al., "A framework for few-shot language model evaluation," Zenodo, 2021, doi: 10.5281/zenodo.5371628.
- [31] Anthropic, "Claude (Claude Opus 5.5), accessed through Claude Code," large language model, Oct. 2026. [Online]. Available: <https://www.anthropic.com/claude>

Sudharsan S is currently pursuing the B.Tech. degree in Computer Science and Engineering at the SRM Institute of Science and Technology, Tiruchirappalli, India. His research interests include natural language processing, large language model evaluation, and variance decomposition methods.

S. Kanaga Suba Raja received the B.E. and M.E. degrees in Computer Science and Engineering. He is currently an Assistant Professor with the Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli, India, and a Visiting Researcher with the Research Institute of IoT Cybersecurity, National Kaohsiung University of Science and Technology, Kaohsiung, Taiwan. His research interests include machine learning, natural language processing, and statistical methods for AI evaluation.

Shree Harish V received the degree from the SRM Institute of Science and Technology, Tiruchirappalli, India. He is currently with the Department of Computer Science and Engineering (Big Data Analytics), SRM Institute of Science and Technology, Tiruchirappalli, India. His research interests include cybersecurity, IoT, and AI systems.

Chin-Shiuh Shieh received the Ph.D. degree in Electrical Engineering. He is currently with the Research Institute of IoT Cybersecurity, Department of Electrical Engineering, National Kaohsiung University of Science and Technology, Kaohsiung, Taiwan. His research interests include wireless networks, cybersecurity, and IoT systems.

Mong-Fong Horng received the Ph.D. degree in Electrical Engineering. He is currently with the Research Institute of IoT Cybersecurity, Department of Electrical Engineering, National Kaohsiung University of Science and Technology, Kaohsiung, Taiwan. His research interests include computer networks, cybersecurity, and AI applications.

Lavanya R received the degree from the SRM Institute of Science and Technology, Tiruchirappalli, India. She is currently with the Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli, India. Her research interests include machine learning, data analytics, and AI systems.

• • •